with a speed of, say, 20 feet per second, when suddenly an unexpected electrical storm stirs overhead, sending you both running for cover. After it passes, you rejoin to resume your game of catch but you notice that something has changed. Your friend's hair has become wild and spiky, and her eyes have grown severe and crazed, When you look at her hand, you are stunned to see that she is no longer planning to play catch with a baseball, but instead is about to toss you a hand grenade. Understandably, your enthusiasm for playing catch diminishes substantially; you turn to run. When your companion throws the grenade, it will still fly toward you, but because you are running, the speed with which it approaches you will be less than 20 feet per second. In fact, common experience tells us that if you can run at, say, 12 feet per second then the hand-grenade will approach you at (20 - 12 =) 8 feet per second. As another example, if you are in the mountains and an avalanche of snow is rumbling toward you, your inclination is to turn and run because this will cause the speed with which the snow approaches you to decrease—and this, generally, is a good thing. Again, a stationary individual perceives the speed of the approaching snow to be greater than that perceived by someone in retreat.

Now, let's compare these basic observations about baseballs, grenades, and avalanches to those about light. To make the comparisons tighter, think about a light beam as composed of tiny "packets" or "bundles" known as photons (a feature of light we will discuss more fully in Chapter 4). When we turn on a flashlight or a laser beam we are, in effect, shooting a stream of photons in whatever direction we point the device. As we did for grenades and avalanches, let's consider how the motion of a photon appears to someone who is moving. Imagine that your crazed friend has swapped her grenade for a powerful laser. If she fires the laser toward you—and if you had the appropriate measuring equipment—you would find that the speed of approach of the photons in the beam is 670 million miles per hour. But what if you run away, as you did when faced with the prospect of playing catch with a hand grenade? What speed will you now measure for the approaching photons? To make things more compelling, imagine that you can hitch a ride on the starship Enterprise and zip away from your friend at, say, 100 million miles per hour. Following the reasoning based on the traditional Newtonian worldview, since you